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ECEN 474/704
Lab 5: Prelab
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1. **Comparison of current-mirror topologies.** Make a table listing the three current-mirror topologies described in this lab. Rate each topology using good, medium, and bad for output resistance, accuracy, complexity, and compliance voltage.

   Solution.

   The comparison assumes comparable device parameters and operating currents. "Good" denotes a desirable property: high output resistance, high accuracy, low complexity, or low required compliance voltage.

Topology	R_{out}	Accuracy	Complexity	Compliance voltage
Simple	Bad	Medium	Good	Good
Cascode	Good	Good	Medium	Bad
Low-voltage cascode	Good	Good	Bad	Medium

   (a) Simple current mirror. The two-transistor mirror has the lowest complexity and requires approximately one overdrive voltage at its output. Its output resistance is approximately r_o. Differences between the reference and output drain voltages produce current error through channel-length modulation.

   (b) Cascode current mirror. The cascode devices reduce changes in the drain voltages of the mirroring devices, improving current accuracy and increasing output resistance to approximately $g_m r_o^2$ when $g_m r_o \gg 1$. For equal overdrives, the conventional topology requires approximately $V_{TH} + 2V_{OV}$ of output voltage.

   (c) Low-voltage cascode current mirror. A separate bias branch sets the cascode gate voltage, allowing high output resistance and good current matching with an ideal minimum compliance voltage near $2V_{OV}$. This is lower than the conventional cascode requirement but higher than that of the simple mirror. The additional bias transistor and bias current increase circuit complexity. Saturation margin increases the practical compliance voltage above the ideal minimum.

2. **Simple 1:1 current-mirror design.** Design a simple 1:1 current mirror with a compliance voltage between 125 and 175 mV, an output current of 120 μA, and an output resistance of at least 25 kΩ. Determine the width and length of each transistor.

Solution.

(a) Process parameters and channel length. The single-finger $W/L = 5/0.27\,\mu$m NFET in the Lab 1 report provides the following extracted parameters:

$$\mu_n C_{ox} = 215.6\,\mu\text{A/V}^2, \quad V_{TH} = 0.433\,\text{V}, \quad \lambda = 0.1624\,\text{V}^{-1}.$$

For $L_1 = L_2 = 0.27\,\mu$m, the first-order output-resistance estimate is

$$R_{\text{out}} \approx \frac{1}{\lambda I_D} = \frac{1}{0.1624(120 \times 10^{-6})} = 51.3\,\text{k}\Omega > 25\,\text{k}\Omega.$$

(b) Overdrive and transistor widths. The center of the specified compliance range gives

$$V_{OV} \approx V_{\text{compliance}} = 0.150\,\text{V}, \qquad V_{GS} \approx V_{TH} + V_{OV} = 0.583\,\text{V}.$$

Neglecting channel-length modulation in the DC current, the square-law model gives

$$I_D \approx \frac{1}{2}\mu_n C_{ox}\frac{W}{L}V_{OV}^2, \qquad \frac{W}{L} = \frac{2(120)}{215.6(0.150)^2} = 49.47.$$

Consequently,

$$\boxed{W_1 = W_2 = 13.36\,\mu\text{m}, \qquad L_1 = L_2 = 0.27\,\mu\text{m}.}$$

With $I_{\text{in}} = 120\,\mu$A, equal dimensions give a nominal 1:1 current ratio. The stated widths are total effective widths, not individual finger widths.

(c) Operating region and interpretation. M1 is diode-connected, with $V_{DS1} = V_{GS1} \approx 0.583\,\text{V} > V_{OV1}$. M2 is in saturation for approximately $V_{\text{out}} \geq 0.150\,\text{V}$ under this model.

The dimensions follow from a first-order model whose extracted parameters depend on geometry and bias. Mobility degradation, characterized by $\theta = 0.9999\,\text{V}^{-1}$ in Lab 1, is omitted from the sizing equation. Channel-length modulation introduces an output-voltage dependence into the drain current. The approximate resistance $1/(\lambda I_D)$ also neglects the finite $(1 + \lambda V_{DS})$ correction. These effects require simulation to determine the actual compliance voltage, output current, and output resistance.

3. **Low-voltage cascode current-mirror design.** Design a low-voltage cascode current mirror with a 1:2 input-to-output current ratio, a 60 μA input current, a low-frequency output resistance greater than 1 MΩ, and a compliance voltage below 500 mV. Determine all transistor widths and lengths and the bias current. Validate saturation at the DC operating point.

Solution.

(a) Topology and design assumptions. In the topology of Fig. 5-3 in the Lab 5 manual, M1 and M2 are the lower mirroring devices; M3 and M4 are their respective upper cascode devices. The drain of M3 connects to the gates of M1 and M2. The diode-connected bias device M_B supplies the gate voltage of M3 and M4. The sources of M1, M2, and M_B connect to V_{SS}; all NFET bulk terminals connect to V_{SS}.

The design uses the Lab 1 parameters of Question 2, with

$$L_1 = L_2 = L_3 = L_4 = L_B = 0.27\,\mu\text{m}, \qquad V_{OV1} = V_{OV2} = V_{OV3} = V_{OV4} = 0.150\,\text{V}.$$

The analysis neglects body effect and mobility degradation. The square law determines the device dimensions, and λ determines the small-signal output-resistance estimate.

(b) Reference and output branch dimensions. For M1 and M3, $I_D = 60\,\mu$A:

$$\left(\frac{W}{L}\right)_{1,3} = \frac{2(60)}{215.6(0.150)^2} = 24.74, \qquad W_1 = W_3 = 6.68\,\mu\text{m}.$$

For M2 and M4, $I_D = 120\,\mu$A:

$$\left(\frac{W}{L}\right)_{2,4} = 49.47, \qquad W_2 = W_4 = 13.36\,\mu\text{m}.$$

The doubled output-branch widths provide the nominal 1:2 current ratio.

(c) Bias branch and compliance voltage. Setting $V_{DS1} \approx V_{DS2} \approx 0.250\,\text{V}$ provides 0.100 V of nominal saturation margin for the lower devices. The required cascode gate voltage is

$$V_B = V_{S3} + V_{GS3} = 0.250 + (0.433 + 0.150) = 0.833\,\text{V}.$$

Thus $V_{OVB} = V_B - V_{TH} = 0.400\,\text{V}$. For $I_{\text{bias}} = 60\,\mu$A, the bias-device dimensions are

$$\left(\frac{W}{L}\right)_B = \frac{2(60)}{215.6(0.400)^2} = 3.479, \qquad \boxed{W_B \approx 0.94\,\mu\text{m}.}$$

The predicted compliance voltage is

$$\boxed{V_{\text{compliance}} \approx V_{S4} + V_{OV4} = 0.250 + 0.150 = 0.400\,\text{V} < 0.500\,\text{V}.}$$

(d) Saturation verification at the DC operating point. At the selected operating point, $V_{\text{out}} = 1.000\,\text{V}$. Ideal current sources supply the input and bias currents. Since the drain of M3 is connected to the gate of M1, $V_{D3} = V_{G1} \approx 0.583\,\text{V}$. The first-order saturation condition is $V_{DS} \geq V_{GS} - V_{TH} = V_{OV}$.

Device	V_{DS} (V)	V_{OV} (V)	Saturation check
M1	0.250	0.150	$0.250 > 0.150$
M2	0.250	0.150	$0.250 > 0.150$
M3	$0.583 - 0.250 = 0.333$	0.150	$0.333 > 0.150$
M4	$1.000 - 0.250 = 0.750$	0.150	$0.750 > 0.150$
M_B	0.833	0.400	$0.833 > 0.400$

All five devices satisfy the saturation condition under the stated first-order assumptions.

(e) Low-frequency output resistance. For the output devices M2 and M4,

$$r_{o2} \approx r_{o4} \approx \frac{1}{0.1624(120 \times 10^{-6})} = 51.3\,\text{k}\Omega, \qquad g_{m4} \approx \frac{2I_D}{V_{OV4}} = 1.60\,\text{mS}.$$

With the cascode gate held at its small-signal bias voltage and neglecting body effect,

$$\begin{aligned} R_{\text{out}} &\approx r_{o2} + r_{o4} + g_{m4} r_{o2} r_{o4} \\ &\approx \boxed{4.32\,\text{M}\Omega > 1\,\text{M}\Omega}. \end{aligned}$$

(f) Design summary and model validity.

Device	Total W (μm)	L (μm)	Nominal I_D (μA)
M1	6.68	0.27	60
M2	13.36	0.27	120
M3	6.68	0.27	60
M4	13.36	0.27	120
M_B	0.94	0.27	60

The nominal bias current is $I_{\text{bias}} = 60\,\mu\text{A}$. Because M3 and M4 have elevated source potentials with their bulks at V_{SS}, body effect raises their thresholds and can reduce the lower devices' saturation margin relative to this calculation. The Lab 1 summary does not provide body-effect parameters, so its effect on the operating point remains unquantified. Simulation of the device operating points determines whether W_B, I_{bias}, or other dimensions require adjustment. The tabulated values therefore represent analytical design estimates.